Argus: Software Requirements Specification (SRS)

National Security Innovation Challenge · Notre Dame Sprint Lab Series · DIU sponsor project

Status key used in this document: *Implemented* means it exists in the current repository. *Planned* means it is in the project plan but not yet in the repository. *Stretch* means we do it only if time allows.

1. Introduction

1.1 Purpose

This document specifies what Argus shall do, how well it shall do it, and how we will check. It is written for the five team members.

1.2 Product scope

Argus lets a person on site with no RF expertise, using a low-cost receive-only radio, find out whether the signals around them are **expected**, **unknown** or **hostile**, in plain English. It does this by capturing a sweep, detecting signals, naming them, judging them against a site profile, and explaining the result in an alert card and a grounded chat.

One shared core serves three use cases (UC):

- **UC1, Is that a drone?** Detect drone control links at 2.4 and 5.8 GHz and tell them apart from Wi-Fi.
- **UC2, Are we being jammed?** Notice a rising noise floor and missing signals.
- **UC3, Is this normal here?** Judge any signal against a site baseline.

Argus does **not** replace EW specialists or counter-drone systems. It gives a first read and says when to call a specialist.

1.3 Definitions

Term	Meaning
SDR	Software-defined radio, here RTL-SDR v4 and HackRF One
IQ	Raw in-phase and quadrature radio samples
Sweep	A power-versus-frequency scan (for example from rtl_power or hackrf_sweep)
Detection	One signal found in a capture, with frequency, bandwidth and power
Laya	The fine-tuned classifier model that picks an answer for each observation
Knowledge file	JSON holding the question asked about each signal and a description of each answer

Term	Meaning
Site profile	A description of what is normal at one site (campus, power plant, field-forward)
Disposition	Blue (expected here), White (unknown), Red (not expected)
EW	Electronic warfare
Remote ID	The broadcast identification that US drones are required to send

1.4 References

- Argus repository: README.md, documentation/document.md, finetune/README.md, documents/local-setup.md
- Argus NSIC pitch deck (first draft) and project plan (74 tasks, 162 story points)
- Gen. Gregory Guillot, NORAD/USNORTHCOM, Senate Armed Services Committee, Feb 13, 2025 (via DefenseScoop), source of the 350 drone detections figure
- Data sources listed in section 8

2. Overall description

2.1 Product perspective

Argus is a standalone, single-machine tool. A person loads or captures signal data, and Argus produces per-signal verdicts and a chat that answers questions about them. It does not transmit, does not decode message content, and does not need a cloud service to run.

The current repository is the **classification and chat core** built at SprintHack@ND by Gabriel Marques and José Lima. The detection, judging and explanation layers described in the pitch are the work of this sprint series.

2.2 Architecture summary

Component	Address	Job	Status
Next.js app	localhost:3000	Library (files) and chat page; forwards uploads	Implemented
LangGraph agent	localhost:2024	Builds each chat turn	Implemented
Laya service (FastAPI)	localhost:8000	Validates files, runs classifier, holds current result	Implemented
Ollama	localhost:11434	Runs the chat model (default qwen2.5:14b)	Implemented
Ingest and detection	n/a	Parse sweeps and IQ, find peaks	Planned
Judge	n/a	Site profile to blue/white/red with reason	Planned

2.3 Users

User	Needs
On-site operator (guard, plant operator, small-unit member)	A color, a plain explanation, and what to do, with no RF knowledge
Site owner or security lead	A site profile that matches their site and a report they can hand on
EW or RF specialist (escalation)	Detection data, confidence and the reasoning behind each verdict
Team and evaluators	Repeatable tests, measured results and clear limits

2.4 Operating environment

- One laptop-class machine, Python 3.12, Node 22, Ollama running. Laya uses CUDA when at least 2 GiB of GPU memory is free, otherwise Apple MPS, otherwise CPU.
- Hardware: 2x RTL-SDR v4 and 1x HackRF One, receive-only. HackRF One tops out at 6 GHz.
- A fine-tuned checkpoint (about 800 MB) in models/laya-rf-modes, not committed to the repository.

2.5 Constraints

- **C-1:** Receive-only. The system never transmits and never generates jamming.
- **C-2:** Only open tools, open protocols and publicly available data. Military frequency plans are not public, so civilian bands act as stand-ins and a site's real assignments plug in as configuration.
- **C-3:** Laya and the chat model share a small GPU and take turns. The service and the agent must name the same chat model.
- **C-4:** The chat model must support tool calling.
- **C-5:** Final demo to DIU on December 5, with feature freeze on December 1.
- **C-6:** Dataset licences must be confirmed before anything built on them is shared.

2.6 Assumptions and dependencies

- Teammates have access to the SDR hardware and a campus location for captures.
- Public datasets and remote receivers stay available.
- DIU can supply or point us to drone and jamming recordings, because we cannot record them ourselves.
- A cloud chat model is allowed for the demo only, with a local model as the target.

3. System features and functional requirements

Priority: **M** must, **S** should, **C** could.

3.1 Capture and ingest

ID	Requirement	Pri	Status
FR-1	The system shall accept a data file as a JSON list or JSONL, each row having center_frequency_hz, bandwidth_hz and modulation ("unknown" allowed).	M	Implemented
FR-2	The system shall accept a knowledge file of one or more questions, each with type "choice" and at least two answers with descriptions.	M	Implemented
FR-3	The system shall refuse an invalid file with a message naming the file and the row or question at fault.	M	Implemented
FR-4	The system shall limit each file to 2 MB.	M	Implemented
FR-5	The system shall parse rtl_power and hackrf_sweep output into a power-versus-frequency table.	M	Planned
FR-6	The system shall ingest IQ recordings in .cu8, .cs8 and SigMF formats.	M	Planned
FR-7	The system shall support a live mode that processes new sweeps as they arrive.	C	Planned

3.2 Detection

ID	Requirement	Pri	Status
FR-8	The system shall find each signal in a sweep or IQ capture and report center frequency, bandwidth, power and the local noise floor.	M	Planned
FR-9	The system shall output detections in the observation format of FR-1 so they feed the classifier unchanged.	M	Planned
FR-10	The system shall report burst timing or duty cycle for each detection so hopping can be told from continuous signals.	S	Planned

3.3 Classification

ID	Requirement	Pri	Status
FR-11	The system shall run Laya on each observation and return one answer with a confidence.	M	Implemented
FR-12	The system shall classify the first 100 rows of a data file per run and classify further slices on request.	M	Implemented

ID	Requirement	Pri	Status
FR-13	The system shall unload the chat model before a run and reload it after, so Laya and the chat model can share one GPU.	M	Implemented
FR-14	The system shall return top-3 candidates with confidence for each detection from a signal-library matcher.	M	Planned
FR-15	The rule-based library matcher shall work on its own if Laya does not transfer to a band.	M	Planned
FR-16	The system shall describe each signal type in plain English from the signal library.	S	Planned

3.4 Judging against a site

ID	Requirement	Pri	Status
FR-17	The system shall load a site profile (campus, power plant, field-forward) describing which signals are expected.	M	Planned
FR-18	The system shall give every detection a disposition of blue, white or red, with a one-line reason.	M	Planned
FR-19	The system shall never assign blue to a detection whose confidence is below the configured threshold; such detections shall be white.	M	Planned
FR-20	The system shall flag a possible jamming event when the noise floor rises and expected signals go missing relative to the site profile.	M	Planned
FR-21	The system shall flag a possible drone control link when short hopping bursts appear in the 2.4 or 5.8 GHz bands that the site profile has not shown.	M	Planned
FR-22	The system should use the FCC frequency allocation table so each explanation can say what a band is meant for.	S	Planned
FR-23	The system could use licensed-transmitter data (FCC ULS) and map data to build or refine a site profile.	C	Stretch
FR-24	The system could check whether a strong signal is plausibly from a known licensed transmitter using terrain and propagation.	C	Stretch

3.5 Explanation and chat

ID	Requirement	Pri	Status

ID	Requirement	Pri	Status
FR-25	The chat shall answer only from the current detections, the site profile and the signal library.	M	Implemented in part (classification results only)
FR-26	The service shall compute counts and statistics and give the chat model a summary, not raw rows.	M	Implemented
FR-27	On every message the agent shall fetch the current result so a new run is picked up.	M	Implemented
FR-28	If nothing has been run or the service is down, the chat shall say so and not invent signals.	M	Implemented
FR-29	The chat shall include confidence and cite the detection it used in each answer.	S	Planned
FR-30	The system shall show an alert card per detection with: what it likely is, why, what to do, and confidence.	M	Planned
FR-31	The system shall provide a glossary of about 40 terms and a define_term tool for the chat.	S	Planned
FR-32	The system shall provide a survey view and a waterfall chart.	S	Planned
FR-33	The system shall export a report of the detections and verdicts.	C	Planned

3.6 Operation

ID	Requirement	Pri	Status
FR-34	The system shall keep one result at a time in memory and refuse a second run started while one is in progress.	M	Implemented
FR-35	The system shall let the operator clear the current result, and shall drop it when a source file is removed.	M	Implemented
FR-36	The system shall offer a command-line run that prints each observation's answer and confidence.	S	Implemented
FR-37	The system shall switch between a cloud chat model and a local one by configuration only.	M	Planned

4. External interface requirements

4.1 User interface

- A Library for adding knowledge and data files, a Run control, and a chat. The chat is paused while the classifier runs.
- Alert cards use the three colors with text labels, so color is never the only signal.
- Language is plain English. Technical terms are explained or defined by the glossary.

4.2 Software interfaces (Laya service)

Request	Purpose
POST /classify	Take knowledge and data (single or lists of files) and classify the first 100 rows
POST /reclassify	Re-run on the stored files with a count or the remaining rows
GET /context	Return the current result as text for the chat model, and whether a run is in progress
DELETE /context	Forget the current result and stored files
GET /health	Return {"status": "ok"}

Planned additions: an endpoint to ingest a sweep or IQ file, an endpoint to load a site profile, and the disposition fields in /context.

4.3 Hardware interfaces

- RTL-SDR v4 and HackRF One accessed through rtl_power and hackrf_sweep, or from recorded files.
- All hardware use is receive-only.

4.4 Data interfaces

- Data file: list of objects with center_frequency_hz, bandwidth_hz (numbers in Hz), modulation (text), optional id. Other fields are ignored today; snr_db and duty_cycle exist in the test data but the classifier does not use them yet.
- Knowledge file: one JSON object of questions, each with type "choice", instructions and criteria.
- Planned: site profile file and signal library file, with schemas merged in Sprint 1.

5. Non-functional requirements

5.1 Performance

- **NFR-1:** Classifying 100 observations shall take about 5 seconds on a 4 GB GPU (measured today). Reloading the chat model takes 4 to 30 seconds.
- **NFR-2:** The system shall show progress while a run is active and shall not accept chat during it.

- **NFR-3 (target):** A sweep-to-verdict run for one capture should finish in under one minute on the demo laptop. This target is not yet measured.

5.2 Reliability and safety of verdicts

- **NFR-4:** A wrong "safe" verdict is the worst failure. Low confidence shall always rank as white, never blue.
- **NFR-5:** The assistant shall not invent a signal. Target: zero invented signals across 30 scripted questions.
- **NFR-6:** The demo shall run end to end three times in a row on the demo laptop, with a recorded fallback available.

5.3 Security and privacy

- **NFR-7:** The system shall measure signals (frequency, power, pattern) and shall not decode content.
- **NFR-8:** In the final configuration all processing, including the chat model, shall run locally. Cloud use is for the demo only and shall be disclosed.
- **NFR-9:** Captures and site profiles shall not leave the machine unless the operator chooses to export.

5.4 Usability

- **NFR-10:** A person with no RF background shall act on an alert. Target: five non-RF students each complete five tasks, one per use case.

5.5 Maintainability

- **NFR-11:** Pipeline code shall have pytest coverage and pass CI (pytest and oxlint) before merge.
- **NFR-12:** Each feature shall live on its own branch and update its technical docs in the same branch.

5.6 Legal and ethical

- **NFR-13:** Argus shall never transmit and shall never generate jamming. Any transmit test happens only in a shielded setup.
- **NFR-14:** Every dataset shall have its licence recorded. The Panoradio HF and sigidwiki licences are not yet confirmed.
- **NFR-15:** Staged drone and jamming scenarios shall come from licensed public recordings or sponsor-provided samples.

6. Data requirements

6.1 Current data

Data	Source	Notes
Training, 8 shortwave modes	Panoradio HF	About 6,200 rows. Bandwidth is measured, but frequency is assigned by the converter. Rows are

Data	Source	Notes
		flagged simulated.
Training, other class	Artemis-DB (sigidwiki export)	About 13,300 rows drawn from catalogue entries, grouped in 11 families. Flagged simulated.
Test set	Held out from both	774 rows, 9 answers

6.2 Known data limits

- The reported 91.6% accuracy is on synthetic or converted data, against 82.7% for a nearest-neighbour lookup on the same three fields. It is not evidence of recognition off the air.
- Jamming and interference families exist in the training data but are folded into one "other" answer today.
- The model sees three text fields. It has not been validated on real over-the-air captures.

6.3 Required new data

Need	Candidate source	Feeds
Real labeled captures on campus	Our own SDRs, known emitters	FR-5 to FR-9, evaluation
Drone examples	DroneRF (Mendeley Data f4c2b4n755), DroneDetect, Remote ID captures	FR-21, UC1
Band purposes	NTIA and FCC frequency allocation tables	FR-22
Nearby licensed transmitters	FCC ULS	FR-17, FR-23
Site context	OpenStreetMap or Overture Maps	FR-23
Jamming reference	OpenSky or ADS-B Exchange (NIC/NACp)	FR-20
Known ISM devices	rtl_433 device list	FR-14, FR-16

7. Verification and acceptance

ID	Acceptance test	Pass criterion	Covers
AT-1	Detection on three staged scenarios (drone, jamming, normal)	Drone and jamming alerts fire on all staged scenarios	FR-18, FR-20, FR-21
AT-2	False-alarm rate on a normal campus survey	Rate measured and published	FR-18, FR-19
AT-3	Usability, 5 non-RF students x 5	Tasks completed with the alert card and	NFR-10

ID	Acceptance test	Pass criterion	Covers
	tasks	chat	
AT-4	Grounding, 30 scripted questions	0 invented signals	FR-25, FR-28, NFR-5
AT-5	Real-capture accuracy	One honest accuracy number on labeled real captures, reported next to the synthetic one	FR-8, FR-11
AT-6	Low-confidence rule	No detection below threshold is blue	FR-19, NFR-4
AT-7	Demo rehearsal	Three clean runs in a row	NFR-6
AT-8	Regression	pytest and oxlint pass in CI	NFR-11

8. Traceability to use cases

Use case	Main requirements
UC1, drone	FR-8, FR-10, FR-14, FR-18, FR-21, FR-30
UC2, jamming	FR-8, FR-18, FR-20, FR-30
UC3, site baseline	FR-17, FR-18, FR-19, FR-22, FR-23, FR-30
All	FR-1 to FR-4, FR-11, FR-25 to FR-29

9. Schedule alignment

Date	Milestone	Requirements in scope
Nov 13	M1: sweep to verdict end to end	FR-5, FR-8, FR-9, FR-11, FR-18, FR-19
Nov 20	M2: campus survey demo and accuracy report	FR-14, FR-17, FR-29, FR-30, FR-31, AT-5
Dec 1	M3: feature freeze, user tests done	FR-20, FR-21, FR-32, AT-1 to AT-4
Dec 5	Final DIU demo, all three use cases	All Must items, AT-7

10. Risks that affect requirements

Risk	Effect on requirements	Response
Laya does not transfer to new bands	FR-11 may not cover 2.4 and 5.8	FR-15: rule-based matcher ships on its own

Risk	Effect on requirements	Response
	GHz	
Peak detection is noisy on real captures	FR-8 quality	Tune on about 150 labeled detections; fixed channel plans as fallback
Real modulation is unknown from a sweep	The classifier leans on the modulation field	Test with modulation set to "unknown" and measure the drop
Operator over-trusts a wrong "safe" verdict	NFR-4	Show confidence; low confidence is white
Red signals cannot be recorded on campus	AT-1	Inject licensed public recordings; ask DIU for samples
Demo date moved from Dec 11 to Dec 5	Schedule	Freeze Dec 1; move live mode and IQ ingest earlier

11. Open issues

- **OI-1:** The repository has no detection, Judge, site profile or disposition code. Confirm whether any of it exists on another branch.
- **OI-2:** Confirm Panoradio HF and sigidwiki licences, and record the licence for each new dataset.
- **OI-3:** Decide which questions Laya answers: keep the single 9-answer shortwave question, add a family question, or add a threat question. The current token budget fits about 12 options in one question.
- **OI-4:** Decide whether duty cycle and SNR are added to the text the classifier reads. This requires retraining.
- **OI-5:** Choose the first two site profiles beyond the campus and who writes them.
- **OI-6:** Agree the confidence threshold for the blue/white boundary.
- **OI-7:** Decide the demo's chat model (cloud for the demo only, or local from the start).